五. 路径积分

1. 数学准备：

Gauss积分： $\int_{-\infty}^{\infty} dx\, e^{-\frac{x^2}{2}} = \sqrt{2\pi} \;\Rightarrow\; \int_{-\infty}^{\infty} dx\, e^{-\frac{ax^2}{2}} = \sqrt{\frac{2\pi}{a}}$

n次借助： $\int_{-\infty}^{\infty} dx\, x^{2n} e^{-\frac{ax^2}{2}} = \sqrt{\frac{2\pi}{a}} \cdot \frac{1}{a^n} \cdot (2n-1)(2n-3)\cdots\cdots 1 = \sqrt{\frac{2\pi}{a}} \frac{1}{a^n}(2n-1)!!$

$\downarrow$

$\langle x^{2n} \rangle = \frac{1}{a^n}(2n-1)!!$ Wick 收缩

含源： $\int_{-\infty}^{\infty} dx\, e^{-\frac{ax^2}{2} + Jx} = \sqrt{\frac{2\pi}{a}}\, e^{J^2/2a}$ $(J \in \mathbb{C})$

$\int_{-\infty}^{\infty} dx\, e^{i(-\frac{ax^2}{2} + Jx)} = \sqrt{\frac{2\pi i}{a}}\, e^{J^2/2ai}$

重积分： $\int d^n x\, e^{-\frac{1}{2}K_{ij}x^i x^j} \longrightarrow y = Sx$，使 $x^T K x = y^T (S^{-1})^T K S^{-1} y = y^T y$ $(K = S^T S$，则 $\det S = \sqrt{\det K})$

$= \int \frac{d^n y}{\det S} e^{-\frac{1}{2}y_i y^i} = \frac{1}{\sqrt{\det K}} \prod_i \sqrt{2\pi} = \sqrt{\frac{(2\pi)^n}{\det K}}$

含源 $\int d^n x\, e^{-\frac{1}{2}K_{ij}x^i x^j + J_i x^i} = \sqrt{\frac{(2\pi)^n}{\det K}}\, e^{\frac{1}{2}J^T K^{-1} J}$

$\int d^n x\, e^{i(-\frac{1}{2}K_{ij}x^i x^j + J_i x^i)} = \sqrt{\frac{(2\pi i)^n}{\det K}}\, e^{\frac{1}{2i}J^T K^{-1} J}$

泛函基底： $\xi^n \to \int dx\, \xi^n$ 扩展 ?]

泛函级数： $F[\xi] = \sum_n f_n \cdot \xi^n$

$\llcorner$ $e^{\xi^2} = \sum_n \frac{1}{n!}\xi^{2n} \to \int dx\, e^{\xi^2} = \sum_n \frac{1}{n!} \underbrace{\int dx\, \xi^{2n}}_{?} =$?]

泛函积分

泛函： $F[\xi] := \mathcal{F} \to \mathbb{C}$， $\to$ 多元泛函 $G[\xi, \eta]$ ……

将ξ理解为随机变量是否可行？]
(是否更易懂？)

eg：定积分 $F[\xi] = \int dx\, \xi(x)$

泛函积分： $\int D\xi\, F[\xi] \equiv \underbrace{\lim_{\Delta x \to 0}}_{\text{可略去不写}} \int \left(\prod_x \frac{d\xi}{c}\right) F(\xi_x)$， $C = C(\Delta x)$ 使积分有限

有一个常数因子的任意性 ?] 哪个，C？

eg： $\int Dq\, \varphi(q) = \iiint dq_1 dq_2 \cdots \varphi(q)$

平移稳定性： $\int D\xi\, F[\xi] = \int D\xi\, F[\xi + \eta]$

(可看作 $\int d^n x$，$n \to \infty$ 的行为 (有C可使结果有限))

Gauss泛函积分

$\int D\xi\, \underbrace{e^{-\frac{1}{2}\xi^2}}_{\text{一种泛函}} = \int D\xi\, e^{-\frac{1}{2}\int dx\, \xi^2(x)} = \int \prod_x \frac{d\xi_x}{c}\, e^{-\frac{1}{2}\sum_x \Delta x\, \xi_x^2} = \prod_x \int \frac{d\xi_x}{c} e^{-\frac{1}{2}\Delta x\, \xi_x^2} = \frac{1}{c}\prod_x \sqrt{\frac{2\pi}{\Delta x}} \to C = \prod_x \sqrt{\frac{2\pi}{\Delta x}}$

平移不变性 $\rightarrow \int D\xi\, e^{-\frac{1}{2}\xi^2 - J\xi} = e^{\frac{J^2}{2}}$, $J\in\mathbb{C}$

厄米核 $\rightarrow \int D\xi\, e^{-\frac{1}{2}\xi K\xi - J\xi} = \frac{1}{\sqrt{\det K}}\, e^{\frac{1}{2}JK^{-1}J}$

$$\int D\xi^{\dagger} D\xi\, e^{-\xi^{\dagger}K\xi + J^{\dagger}\xi + J\xi^{\dagger}} = \frac{1}{\det K}\, e^{J^{\dagger}K^{-1}J}$$

$$= \int \prod_x\prod_y \frac{d\xi\, d\xi^*}{c^2}\, e^{-\frac{1}{2}\int \xi^{\dagger}_{(x)}K(x,y)\,\xi_{(y)} + J\xi^{\dagger}_{(x)} + J\xi_{(y)}\, dx\, dy}$$

$$= \prod_{x,y}\int \frac{d\xi\, d\xi^*}{c^2}\, e^{\Delta x\Delta y\cdot\underline{(-\xi^{*}_x K\xi_y + J\xi^{*}_x + J\xi_y)}} \rightarrow -(\xi^{*}_x - J^{\dagger}K^{-1})K(\xi_y - K^{-1}J) + J^{\dagger}K^{-1}J$$

$$= \underline{\prod \frac{1}{c}\sqrt{\frac{2\pi}{\Delta x}}}\cdot\frac{1}{\det K}\cdot e^{J^{\dagger}K^{-1}J}$$

c 归一化

泛函微商：y点处对 $\xi(x)$ 改变 $\varepsilon\delta(x-y)$

$$\frac{\delta F[\xi(x)]}{\delta\xi(y)} = \lim_{\varepsilon\to 0}\frac{F[\xi(x)+\varepsilon\delta(x-y)] - F[\xi(x)]}{\varepsilon} \longrightarrow \frac{\delta}{\delta\xi(y)}\xi(x) = \delta(x-y)$$

eg. 简写：$\xi = \int dx\, \xi(x)$ ， $\xi\eta = \int dx\, \xi(x)\eta(x)$ ， $\xi G\eta = \int dx\,dy\, \xi(x)G(x,y)\eta(y)$

有：$\frac{\delta}{\delta\xi(x)}\xi = \int dy\, \delta(x-y) = 1$ ； $\frac{\delta}{\delta\xi(x)}\xi G\eta = \frac{\int dy\,dz\, \varepsilon(x-y)G(y,z)\eta(z)}{\varepsilon} = \int dz\, G(x,z)\eta(z) = G\eta$

$\frac{\delta}{\delta\xi(x)}e^{-\frac{1}{2}\xi^2} = -\xi(x)e^{-\frac{1}{2}\xi^2}$ ； $\frac{\delta}{\delta\xi(x)}e^{-\xi\eta} = -\eta e^{-\xi\eta}$

量子场论中心等式：$\int D\xi\, F(\xi)\, e^{-\frac{1}{2}\xi K\xi + J\xi} = \int D\xi \sum f_n\xi^n e^{-\frac{1}{2}\xi K\xi + J\xi} = \sum_n f_n\int D\xi\, \xi^n e^{-\frac{1}{2}\xi K\xi + J\xi} = \sum_n f_n\int D\xi \left(\frac{\delta}{\delta J}\right)^n e^{-\frac{1}{2}\xi K\xi + J\xi}$

$$= \sum_n f_n\left(\frac{\delta}{\delta J}\right)^n\int D\xi\, e^{-\frac{1}{2}\xi K\xi + J\xi} = F(\delta/\delta J)\frac{e^{\frac{1}{2}JK^{-1}J}}{\sqrt{\det K}}$$

$\Downarrow$

$$\int D\xi\, e^{-\frac{1}{2}\xi K\xi - V(\xi) + J\xi} = e^{-V(\delta/\delta J)}\, e^{\frac{1}{2}JK^{-1}J}$$

(略去 $\frac{1}{\sqrt{\det K}}$)

Grassmann 变量

乘法次序不可换 → q数 （也叫Grassmann代数）

n维Grassmann生成元 ξ_i，满足：$\{\xi_i, \xi_j\} = 0$ （反对易） → $\xi_i^2 = 0$ → 展开式仅有有限项

Grassmann代数拓展口

$f(\xi) = p + q\xi$ ；$g(\eta, \xi) = p + q\xi + r\eta + s\eta\xi$ 或 $-s\xi\eta$ （p.q.r.s为c数(可对易数)）

共轭：$(\xi^*)^* = \xi$ ；$(\xi\eta)^* = \eta^*\xi^*$

积分/微分：

平移不变性 $\int d\xi f(\xi+\eta) = \int d\xi f(\xi)$ 　$f(\xi+\eta) = p + q\xi + q\eta$，$\int d\xi\, q\eta = 0$ 对 $\forall q\ \forall \eta$ 成立 → $\int d\xi = 0$

$\int d\xi\,\xi$ 看作 $\int d\xi$ 与 ξ 之积，故 $\{\eta, \int d\xi\,\xi\} = 0$ → $\int d\xi\,\xi$ 可视为c数 → 定义 $\int d\xi\,\xi = 1$

$$\int d\xi\, g(\xi,\eta) = \int d\xi (p + q\xi + r\eta + s\xi\eta) = q + s\eta$$

颇有逆变与协变的味道

$$\int d\xi\, f(\xi) = \int d\xi\, q\xi = \begin{cases} q & q\text{为c数} \\ -q & q\text{为q数} \end{cases}$$

n维：$\xi = (\xi_1 \cdots \xi_n)$，线性变换 $\xi = S\eta$ —— 要求：$\int d\xi\xi = 1$，$\int d\xi\eta = 0$ 　即：$\int d\xi_1 d\xi_2 \cdots d\xi_n\, \xi_1\xi_2\cdots\xi_n = \int d\eta_1 d\eta_2 \cdots d\eta_n\, \eta_1\eta_2\cdots\eta_n$

$\prod_i \xi_i = (\det S)\prod_i \eta$ → $d^n\xi = \frac{1}{\det S} d^n\eta$ （P数与C数微分规则不同）

微商：（含乘法 $\frac{\partial}{\partial\xi_i}$，故注意次序）

$$\frac{\partial g(\xi,\eta)}{\partial\xi} = q + s\eta \quad , \quad \frac{\partial g}{\partial\eta} = r - s\xi$$

$$\{\xi_i, \frac{\partial}{\partial\xi_j}\} = \delta_{ij} \quad . \quad \{\frac{\partial}{\partial\xi_i}, \frac{\partial}{\partial\xi_j}\} = 0$$

Guass积分：考虑 ξ 与 $\bar\xi$，有：$e^{-\bar\xi\xi} = 1 - \bar\xi\xi + 0$

$$\int d\bar\xi d\xi\, e^{-\bar\xi\xi} = \int d\bar\xi d\xi + \int d\bar\xi d\xi\,\xi\bar\xi = 1 \ , \quad \int d\bar\xi d\xi\, e^{-\bar\xi a\xi} = \int d\bar\xi d\xi\,\xi a\bar\xi = \int d\bar\xi\, a\bar\xi = a = e^{\ln a}$$

n维 $\xi = (\xi_1, \cdots \xi_n)$，$\bar\xi = (\bar\xi_1, \cdots \bar\xi_n)$

$$\int d^n\bar\xi d^n\xi\, e^{-\bar\xi\xi} = (-1)^{(n-1)!}\int \Big(\prod_i d\bar\xi_i d\xi_i\Big) e^{-\sum \bar\xi_i\xi_i} = (-1)^{(n-1)!}\prod_i \int d\bar\xi_i d\xi_i\, e^{-\bar\xi_i\xi_i} = (-1)^{(n-1)!}\prod_i \int d\bar\xi_i (-\bar\xi_i) d\xi_i\,\xi_i = 1$$

n一般取∞，故n!/2为1

变换：$\xi = S\eta$，$\bar\xi = \bar S\bar\eta$ → $\int d^n\bar\xi d^n\xi\, e^{-\bar\xi\xi} = \frac{1}{\det K}\int d^n\bar\eta d^n\eta\, e^{-\bar\eta^T \bar S^T S\eta} = 1$ ⇒ $\int d\bar\eta d\eta\, e^{-\bar\eta K\eta} = \det K$

$(K = \bar S^T S)$

故 $\int D\bar\eta D\eta\, e^{-\bar\eta K\eta} = \det K$

2. 路径积分

跃迁振幅：$\psi(q,t) = \langle q|\psi(t)\rangle = \langle q|U(t,t_0)|\psi(t_0)\rangle = \int dq_0\, K(q,t;q_0,t_0)\psi(q_0,t_0)$

$K(q,t;q_0,t_0) = \langle q|U(t,t_0)|q_0\rangle \rightarrow$ 跃迁振幅

路径积分：$U(t,t_0) = e^{-iH(t-t_0)}$，$\Delta t = \frac{t-t_0}{n}$ 有：$U(t,t_0) = e^{-iH(n\Delta t)} = (1-i\Delta t H)^n$

↓ 代入跃迁振幅

$$K(q,t;q_0,t_0) = \int dq_1 \cdots \int dq_{n-1} \langle q|(1-i\Delta t H)|q_{n-1}\rangle \cdots\cdots \langle q_1|1-i\Delta t H|q_0\rangle$$

每个因子 $\langle q_{m+1}|1-i\Delta t H|q_m\rangle = \int dP_m \underbrace{\langle q_{m+1}|P_m\rangle}_{\frac{1}{\sqrt{2\pi}}e^{iP_m q_{m+1}}} \underbrace{\langle P_m|1-i\Delta t H|q_m\rangle}_{\frac{1}{\sqrt{2\pi}}e^{-iP_m q_m}(1-i\Delta t H(P_m,q_m))}$

$$= \int \frac{dP_m}{2\pi} e^{iP_m(q_{m+1}-q_m)} e^{-iH(P,q)\Delta t}$$

$\downarrow n\to\infty,\ \Delta t\to 0$

$$K(q,t;q_0,t_0) = \int Dq\,Dp\; e^{i\int dt\, p\dot{q} - H(p,q)} \qquad \int Dq = \lim_{\substack{\Delta t\to 0\\ n\to\infty}} \prod_{m=1}^{n-1} \int \frac{dq(t_m)}{\sqrt{2\pi}} \qquad \int Dp = \lim_{\substack{\Delta t\to 0\\ n\to\infty}} \prod_{m=1}^{n-1} \int \frac{dP(t_m)}{\sqrt{2\pi}}$$

Feynman 公式

对 $H = \frac{P^2}{2m} + V(q)$，$K(q,t;q_0,t_0) = \int Dq\, V^{-i\int dt V(q)} \int Dp\, e^{\int dt(-\frac{i}{2m}P^2 + iP\dot{q})}$

对P积分：$\int Dp\, e^{\int dt(-\frac{i}{2m}P^2 + iP\dot{q})} = N e^{\int dt \frac{1}{2}(i\dot{q})(\frac{m}{i})(i\dot{q})} = N e^{i\int dt \frac{1}{2}m\dot{q}^2}$ N为归一化常数

$$K(q,t;q_0,t_0) = N\int Dq\, e^{i\int dt L(q,\dot{q})}$$

$N = \lim_{n\to\infty} \frac{1}{\sqrt{\det(\frac{1}{m})}} \left(\frac{1}{2\pi}\sqrt{\frac{2\pi}{i\Delta t}}\right)^n = \lim_{n\to\infty} \left(\frac{m}{i2\pi\Delta t}\right)^{\frac{n}{2}}$

↓ 更一般 $L(q,\dot{q}) = \frac{1}{2}\dot{q}_i m^{ij} \dot{q}_j + A^i(q)\dot{q}_i - V(q)$ 同样成立

$$K(q,t;q_0,t_0) = N\int Dq\, e^{iS}$$

(相空间体元大小 $V_h = ih$?]

量子作用量原理 所有路径概率和 ?]

$P = \frac{1}{i\hbar}\frac{\partial}{\partial x}$，$x = x \rightarrow [P,x] = -i\hbar = S_h \rightarrow K = N\int Dq\, e^{\frac{S}{S_h}}$?]

路径积分可推出薛方程，故也可看作基本假设

路径积分量子力学

考虑时间间隔 $t \to t+\Delta t$ ，$K(q,t;q_0,t_0) = Ce^{[i\Delta t L(\frac{q+q_0}{2}, \frac{q-q_0}{\Delta t})]} = Ce^{i\Delta t[\frac{m\eta^2}{2(\Delta t)^2} - V(q+\frac{\eta}{2})]}$ ，$\eta = q_0 - q$

$$\psi(q,t+\Delta t) = C\int d\eta\, e^{i\Delta t(\frac{m\eta^2}{2\Delta t^2} - V(q+\frac{\eta}{2}))}\psi(q+\eta, t)$$

$\Delta t \to 0$ 时 $\psi(q,t) + \Delta t\frac{\partial\psi}{\partial t} = C\int d\eta\, \underbrace{e^{\frac{im\eta^2}{2\Delta t}}}_{\eta\to 0\text{ 才有效}}\underbrace{[1 - i\Delta t V(q)]}_{\text{小量展开}}\underbrace{[\psi(q,t) + \eta\frac{\partial\psi}{\partial q} + \cdots]}_{\text{Taylor}}$

$\downarrow$

$C\int d\eta\, e^{\frac{im\eta^2}{2\Delta t}} = 1$ ，$i\hbar\frac{\partial}{\partial t}\psi(q,t) = [-\frac{1}{2m}\frac{\partial^2}{\partial q^2} + V(q)]\psi(q,t)$ $\longrightarrow$ Schrödinger 方程

动能 $-\frac{1}{2m}\frac{\partial^2}{\partial q^2} \longrightarrow$ 动量 $p = i\frac{\partial}{\partial q}$ $\longrightarrow$ 包含正则量子化

故路径积分可作为一量子化

绘景

Schrödinger 绘景 $\frac{\partial}{\partial t}F = 0$ ，$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle$

Heisenberg 绘景 $\frac{\partial}{\partial t}|\psi_H\rangle = 0$ ，$i\hbar\frac{\partial}{\partial t}F_H = [F_H, H]$

Dirac 绘景：$i\frac{\partial}{\partial t}|\psi(t)\rangle_I = H_I'|\psi(t)\rangle_I$ ，$i\frac{\partial}{\partial t}F_I = [F_I, H_0]$ ← F 与 $|\psi\rangle$ 分别承担一部分变化

跃迁振幅的 Huygens 原理

为何相对论下还有含时/不含之分，需改进H？

编时积的作用与物理表现？
(记得补充Huygens定理)

H绘景：H不含时：$K(q,t;q_0,t_0) = \langle q|e^{-iH(t-t_0)}|q_0\rangle = \langle q,t|q_0 t_0\rangle$

设 $t_1 \in (t_0, t)$ ，$\langle q,t|q_0,t_0\rangle = \int dq(t_1)\langle q,t|q(t_1)\rangle\langle q(t_1)|q_0,t_0\rangle$

$|q_0,t_0\rangle$ 先到 $|q(t_1)\rangle$ 再到 $|q,t\rangle$

$$\langle q,t|q_0,t_0\rangle = \int dq(t_1)\int Dq'' e^{i\int_{t_1}^{t}dt L(q'',\dot{q}'')}\int Dq' e^{i\int_{t_0}^{t_1}dt L(q',\dot{q}')}$$
$$= \int Dq\, e^{i\int_{t_0}^{t}dt L(q,\dot{q})}$$

时序乘积

$$\langle q,t|q(t_1)|q_0,t_0\rangle = \int dq(t_1)\langle q,t|q(t_1)\rangle\langle q(t_1)|q(t_1)|q_0,t_0\rangle = \int_{q(t_1)}dq(t_1)\langle q,t|q(t_1)\rangle\langle q(t_1)|q_0,t_0\rangle = \int Dq\, q(t_1)e^{i\int_{t_0}^{t}dt L(q,\dot{q})}$$

展开：$\langle q,t|q(t_1)q(t_2)|q_0,t_0\rangle = \int Dq\, q(t_1)q(t_2)e^{i\int_{t_0}^{t}dt L(q,\dot{q})}$

Huygence 原理：要求 $t_1 > t_2$ → 时序性

($|q_0,t_0\rangle$ 先到 $|q(t_2)\rangle$ 后到 $q(t_1)\rangle$ ，$\int_{t_2}^{t_1}$ 不决定 $q(t_1)$)

$t_0<t_i<t$, $\langle q,t|\mathcal{T}q(t_1)q(t_2)\cdots q(t_n)|q_0,t_0\rangle = \int Dq\, q(t_1)q(t_2)\cdots q(t_n)e^{i\int_{t_0}^{t}dt L(q,\dot q)}$

编时与Dyson展开的关系?]

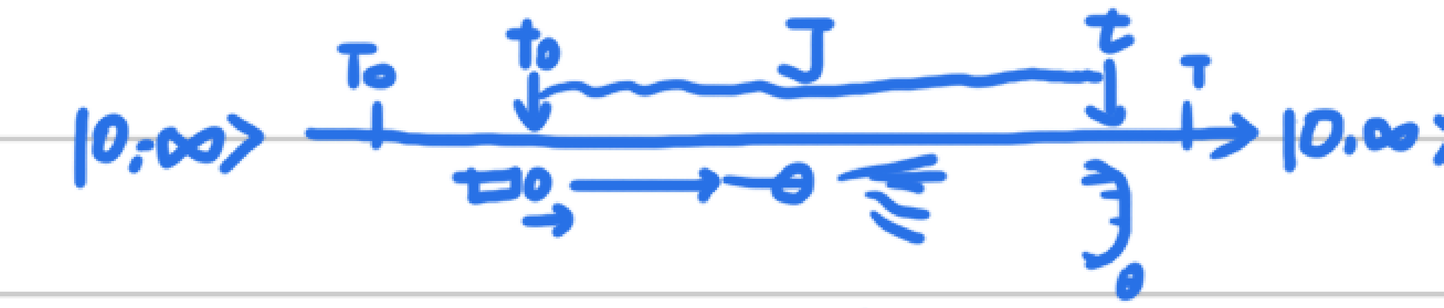
$\mathcal{T}$:编时算符,为$t_1\sim t_n$排序: $t_i>t_j>t_k$

有外源: $L(q,\dot q) \mathrel{+}= Jq \to \langle q,t|q_0,t_0\rangle_J = \int Dq\, e^{i\int dt[L(q,\dot q)+Jq]} = \int Dq\, e^{i\int dt\, Jq}e^{i\int dt L(q,\dot q)}$

$= \langle q,t|\mathcal{T}e^{i\int_{t_0}^{t}dt\, Jq}|q_0,t_0\rangle$

即:对任一态, $\langle\psi,t|\mathcal{T}e^{i\int_{t_0}^{t}dt\, Jq}|\psi_0,t_0\rangle$

($\langle q,t|$完备性: $\langle\psi,t| = \int\langle q,t|\psi,t\rangle dq\langle\psi,t|$)



散射问题必须守护的3件事物:
粒子的拉氏密度
外源J的生成泛函
适合的传播子

散射问题 粒子间的散射与反应

源(S) 相互作用(I) 探测器(D)

$t\to-\infty$ $t\to\infty$

$|in\rangle \longrightarrow \longrightarrow \longrightarrow |out\rangle$

真空态

可用J使粒子在$|0\rangle$中产生,在探测器消失

$\downarrow$

$|0,\infty\rangle$与$|0,-\infty\rangle$为$|out\rangle$与$|in\rangle$

$\downarrow$

J负责中间过程(包括粒子生灭)

Wick转动?]

基态→基态跃迁振幅

这一节内容在孙昌璞量力中第二章重合极多,看来孙书也颇有可取之处

$|in\rangle\to|out\rangle$ $Z[J] = \langle 0,+\infty|0,-\infty\rangle_J$ 为J的泛函 生成泛函

这么做原因?]

$\langle Q,T|n\rangle$

设J有效于 $t_0\sim t$, $T_0<t_0<t<T$

温习孙书?]

H绘景: $\langle Q,T|Q_0,T_0\rangle_J = \int dq\,dq_0\langle Q,T|q,t\rangle\langle q,t|q_0,t_0\rangle_J\langle q_0,t_0|Q_0,T_0\rangle$

$\langle Q,T|q,t\rangle\langle q_0,t_0|Q_0,T_0\rangle = \sum_{m,n}\langle Q,T|m\rangle\langle m|q,t\rangle\langle q_0,t_0|n\rangle\langle n|Q_0,T_0\rangle$

这里是否也可与缓变/快变对应?]

H本征态展开

$= \sum_{m,n}\psi_m^*(Q)\psi_n(Q_0)\cdot\psi_m(q)\psi_n^*(q_0)\, e^{-iE_n(T-t)}\, e^{iE_n(T_0-t_0)}$

$\langle m|q,t\rangle = \psi_m(q,t) = \psi_m(q)e^{iE_mt}$

t延拓到复平面 → 转角$\delta\le\frac{\pi}{2}$ (Wick 转动) → $T_0\to-\infty e^{-i\delta}$, $T\to\infty e^{-i\delta}$ ↓

朗道-齐纳近似?]

$\langle q,t|Q,T\rangle$ 快速衰减

保留衰减最慢项:

$\langle Q,T|q,t\rangle\langle q_0,t_0|Q_0,T_0\rangle = \psi_0^*(Q)\psi_0(Q_0)\psi_0(q,t)\psi_0^*(q_0,t_0)\, e^{-iE_0(T-T_0)}$

有: $\dfrac{\langle Q,T|Q_0,T_0\rangle}{\psi_0^*(Q)\psi_0(Q_0)\, e^{-iE_0(T-T_0)}} = \int dq\,dq_0\, \psi_0(q,t)\langle q,t|q_0,t_0\rangle_J\psi_0^*(q_0,t_0)$

| $t\to\infty$, $t_0\to-\infty$ 时要求 $\langle 0,\infty|0,-\infty\rangle$]

$$N = \lim_{\substack{Q \to +\infty e^{-i\delta} \\ Q_0 \to -\infty e^{-i\delta}}} \psi_0^*(Q)\psi_0(Q_0)e^{-iE_0(T-T_0)} \text{（归一化常数）}$$

生成泛函作用：$Z[J]$ 是 Green 函数 G 的生成函数：

$$G_n = \frac{1}{Z[J]} \cdot \frac{\delta^n Z[J]}{i^n \delta J^n}\Big|_{J=0}$$

$$Z[J] = \langle 0,\infty|0,-\infty\rangle_J = N\int Dq\, e^{i\int_{-\infty e^{-i\delta}}^{\infty e^{-i\delta}} dt[L(q,\dot q) + Jq + \frac{i\varepsilon}{2}q^2]}$$

加1项 $\frac{i\varepsilon}{2}q^2$ 使快速衰减

泛函微商：$Z[J] = \langle 0,\infty|0,-\infty\rangle_J = \langle 0,\infty|\mathcal{T}e^{i\int dt Jq}|0,-\infty\rangle$

$$\frac{\delta^n Z}{i^n \delta J(t_1)\cdots\delta J(t_n)}\Big|_{J=0} = \langle 0,\infty|\mathcal{T}q(t_1)\cdots q(t_n)|0,-\infty\rangle$$

Wick转动 $\delta=\frac{\pi}{2}$ 时 → 纯虚轴 → $\tau = it = x_4$ → $-ds^2 = \sum_{\mu=1}^{4}(dx_\mu)^2$ 注意度规 $[1,-1,-1,-1]$ → $[1,1,1,1]$

x_i 为欧氏空间，$g_{ij} = I_{ij}$

$$Z_E[J] = N\int Dq\, e^{\int_{-\infty}^{\infty} d\tau[L(q, i\frac{dq}{d\tau}) + J(\tau)q(\tau)]}$$

$$= N\int Dq\, e^{-\int_{-\infty}^{\infty} d\tau[L_E(q, \frac{dq}{d\tau}) - J(\tau)q(\tau)]}$$

$L_E(q, \frac{dq}{d\tau}) = -L(q, \frac{i\,dq}{d\tau})$

$L = T-V \to L_E = T+V$ 正定

L_E 正定 → $Z_E[J]$ 收敛（e^{-A^2}）

$$\frac{1}{Z[J]}\frac{\delta^n Z[J]}{i^n \delta J(t_1)\delta\cdots\delta J(t_n)}\Big|_{J=0} = \frac{1}{Z_E[J]}\frac{\delta^n Z_E[J]}{\delta J(\tau_1)\delta J(\tau_2)\cdots\delta J(\tau_n)}\Big|_{J=0}$$

3.标量场路径积分

1)跃迁振幅： $q(t)\to\phi(t)$. $L\,dt\to\mathcal{L}\,d^4x$. $Dq\to D\phi$

约束：$\phi_0=\phi_0(x)$, $\phi=\phi(x)$

$$K(\phi,t;\phi_0,t_0)=N\int D\phi\, e^{i\int d^4x\,\mathcal{L}(\phi,\partial_\mu\phi)}$$ (x取$\mathbb{R}^3$. t取 (t_0,t))

基态 $|\infty,0\rangle=|-\infty,0\rangle=|0\rangle$ 真空态、

2)生成泛函 $Z_0[J]$： $\mathcal{L}=\frac{1}{2}(\partial_\mu\partial^\mu\phi-m^2\phi)$ ，规范变化 $\mathcal{L}+=\partial^\mu A_\mu$ → 可有：$\partial_\mu\phi\partial^\mu\phi\longrightarrow-\phi\partial^2\phi$

加外源，小负虚数项： $\mathcal{L}(\phi,\partial_\mu\phi)=\mathcal{L}_0+J\phi+\frac{i\varepsilon}{2}\phi^2=-\frac{1}{2}\phi(\partial^2+m^2-i\varepsilon)\phi+J\phi$

$$Z_0[J]=N\int D\phi\, e^{-i\int d^4x[\frac{1}{2}\phi(\partial^2+m^2-i\varepsilon)\phi-J\phi]}$$

$$=N'e^{-\frac{1}{2}J[i(\partial^2+m^2-i\varepsilon)]^{-1}J}=N'e^{-\frac{i}{2}J\Delta_F J}$$

$\Delta_F=-(\partial^2+m^2-i\varepsilon)^{-1}$ 格林函数？]

N'有：$Z_0[J]=\langle0,\infty|0,-\infty\rangle_J=\langle0|0\rangle_J$ → $Z_0[0]=1$ → $N'=1$

$$Z_0[J]=N\int D\phi\, e^{-i\int d^4x[\frac{1}{2}\phi(\partial^2+m^2-i\varepsilon)\phi-J\phi]}=e^{-\frac{i}{2}\int dx\,dy\,J_{(x)}\Delta_F(x-y)J_{(y)}}$$
$$=e^{-\frac{i}{2}J\Delta_F J}$$

$$\frac{1}{N}=\int D\phi\, e^{i\int d^4x[\frac{1}{2}\phi(\partial^2+m^2-i\varepsilon)\phi]}$$

3) Feynmann传播子Δ_F ： $\int dx\,(\partial^2+m^2-i\varepsilon)\Delta_F=-1$ → 1) $(\partial^2+m^2-i\varepsilon)\Delta_F=-\delta(x-y)$

E-L方程：$(\partial^2+m^2-i\varepsilon)\phi(x)=J(x)$ ⟶ $\phi(x)=\int dy\,\Delta_F(x-y)J(y)$

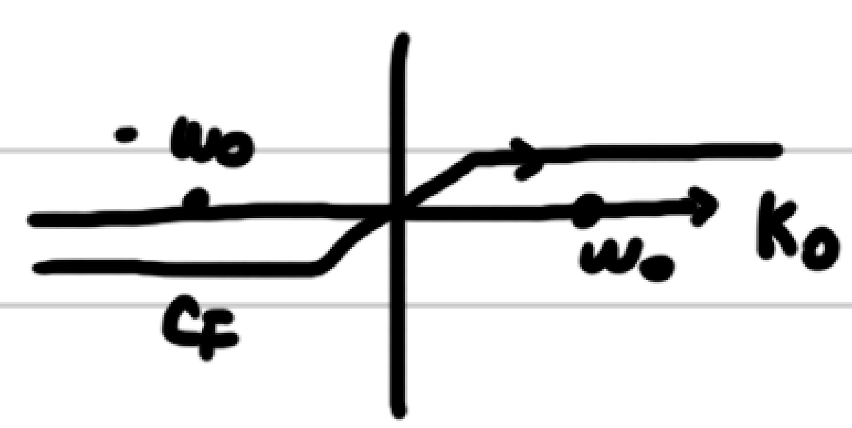
Δ_F为y点源产生的场 $y\to x$

对一个特定问题，可取不同的k_0路径，得到不同传播子(用留数定理，0点共2个，$k_+=\sqrt{m^2-i\varepsilon}$ ，$k_-=-\sqrt{m^2-i\varepsilon}$，故所有传播子都可视作2个$\Delta(\Delta_+,\Delta_-)$与时间限制的线性和)

动量空间：$\Delta_F=\int\frac{d^4k}{(2\pi)^4}\Delta_F(k)e^{-ikx}=\frac{d^4k}{(2\pi)^4}\frac{1}{k^2-m^2+i\varepsilon}e^{-ikx}$ → $\Delta_F(k)=\frac{1}{k^2-m^2+i\varepsilon}$

Δ_F有2个奇点：$k_0=\omega-i\varepsilon$；$k_0=-\omega+i\varepsilon$ ($\varepsilon\to0^+$时在k_0轴上)

避开奇点得路径C_F

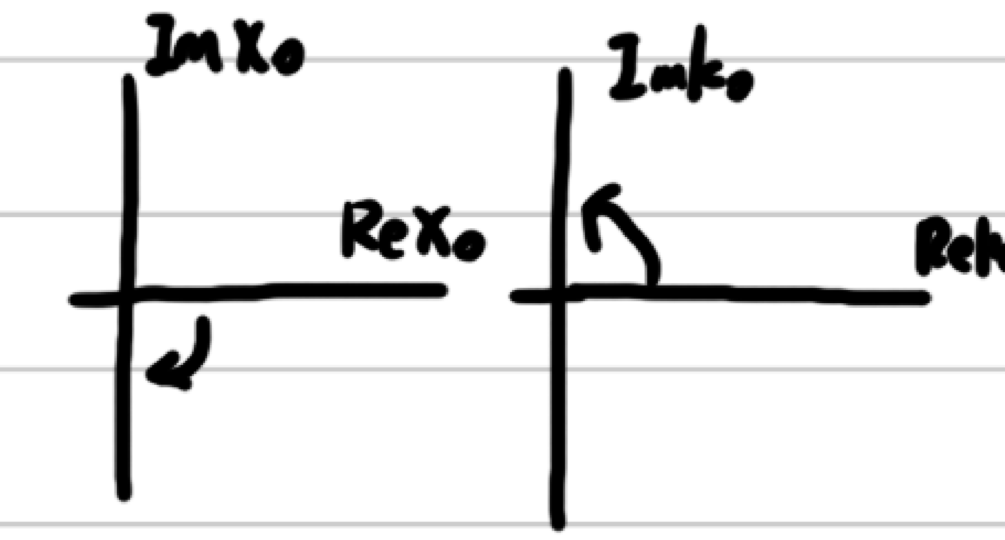


两种方法确定路径：欧氏空间，格林函数

欧氏空间法：$x_E=(\vec{x}, x_4=it)$，$x_E^2=-x^2$，$d^4x_E=id^4x$

$k_E=(\vec{k}, k_4=-ik_0)$，$k_E^2=-k^2$　$d^4k_E=-id^4k$ ⟶ 使t在负虚数轴，k_0在正虚轴　$t=-i\tau$

$k_\mu x^\mu \longrightarrow k_4x_4-\vec{k}\cdot\vec{x}$，$\partial_E^2=-\partial^2$ $(\partial^2=\partial_t^2-\nabla^2 \to \partial_E^2=\nabla^2+\partial_\tau^2)$



$$Z_{0E}[J]=\int D\phi\, e^{-\int d^4x_E[\frac{1}{2}\phi(-\partial_E^2+m^2)\phi-J\phi]}=e^{-\frac{1}{2}J(\partial_E^2-m^2)^{-1}J}=e^{\frac{i}{2}J\Delta_F J}$$

$(\partial_E^2-m^2)\Delta_F=-i\delta(x_E-y_E) \to k_4=\pm i\sqrt{k^2+m^2}$ 不在实轴

故有：积分路径沿k_4

格林函数法

中与J是否如同x, p ？］

定义n点格林函数 $G(x_1\cdots x_n)=\langle 0|T[\phi(x_1)\phi(x_2)\cdots\phi(x_n)]|0\rangle$

$$=\frac{\delta^n Z[J]}{i^n\delta J(x_1)\delta J(x_2)\cdots\delta J(x_n)}\Big|_{J=0}$$

自由场Green函数称为自由Green函数

$$Z[J]=\sum_{n=0}^{\infty}\frac{i^n}{n!}\int dx_1\cdots dx_n\, G(x_1,\cdots,x_n)J(x_1)\cdots J(x_n)$$

1点Green函数： $J\Delta_F J=J_x\Delta_F^{xy}J_y=\int dxdy\, J(x)\Delta_F(x-y)J(y)$

$$G(x)=\langle 0|\phi(x)|0\rangle=\frac{\delta Z_0[J]}{i\delta J(x)}\Big|_{J=0}=\frac{\delta}{i\delta J(x)}e^{-\frac{i}{2}J\Delta_F J}\Big|_{J=0}=-\Delta_{Fxy}J_y e^{-\frac{i}{2}J\Delta_F J}\Big|_{J=0}=0$$

从$\phi(x)$来看，$\phi(x)=\phi^{(+)}(x)+\phi^{(-)}(x)$，→ 场真空期望值为0 （即产生与湮灭必须同时）

a_k,湮灭　$a_k^\dagger$,产生

？］如何理解

2点Green函数： $G(x,y)=\langle 0|T[\phi(x),\phi(y)]|0\rangle$

$$=\frac{\delta^2 e^{-\frac{i}{2}J\Delta_F J}}{-\delta J(x)\delta J(y)}=\frac{\delta}{i\delta J(x)}\left(-J_z\Delta_{Fzy}e^{-\frac{i}{2}J\Delta_F J}\right)=\left[i\Delta_{Fxy}+(-J_z\Delta_{Fzy})(-J_z\Delta_{Fzx})\right]e^{-\frac{i}{2}J\Delta_F J}\Big|_{J=0}$$

与第2章的协变化是否有关

$=i\Delta_F(x-y)$ → Feynmann传播子

$T[\phi(x),\phi(y)]=\theta(x^0-y^0)\phi(x)\phi(y)+\theta(y^0-x^0)\phi(y)\phi(x)$

$$G(x,y)=\theta(x^0-y^0)\underbrace{\langle 0|\varphi^{(+)}_{(x)}\varphi^{(-)}_{(y)}|0\rangle}_{y生x灭}+\theta(y^0-x^0)\underbrace{\langle 0|\varphi^{(+)}_{(x)}\varphi^{(-)}_{(y)}|0\rangle}_{x生y灭}$$

↓ x与y点的传播

n点格林函数：Wick定理

$$3:\ G(x,y,z)=\frac{\delta\frac{\delta^2 Z[J]}{\delta J_x\delta J_y}}{i\delta J(z)}\Big|_{J=0}=\frac{\delta\big[\overbrace{i\Delta_{Fxy}+(-J_z\Delta_{Fzy})(-J_z\Delta_{Fzx})}^{F_{xy}}\big]e^{-\frac{1}{2}J\Delta_F J}}{\delta J(z)}\Big|_{J=0}$$

$$=\Big[-iJ_z\Delta_F\cdot F_{xy}-\Delta_{Fzy}(-J_1\Delta_{F1x})-\Delta_{Fzx}(-J_1\Delta_{F1x})\Big]e^{-\frac{1}{2}J\Delta_F J}\Big|_{J=0}=0$$

$$4:\ G(x_1,x_2,x_3,x_4)=\frac{\delta\frac{\delta^3 Z[J]}{\delta x_{1,2,3}}}{i\delta J(x_4)}=\frac{\delta^4}{i^4\delta^4}e^{-\frac{1}{2}J_xG^{xy}J_y}\Big|_{J=0}=\frac{\delta^3}{i^3\delta^3}(iJ_xG_{xx_4})Z[J]\Big|_{J=0}$$

$$=\frac{\delta^2}{i^2\delta^2}\big(G_{x_3x_4}+(J_xG^x{}_{x_3})(J_xG^x{}_{x_4})\big)Z[J]\Big|_{J=0}$$

$$=\frac{\delta}{i\delta}\big(iG_{x_2x_3}(J_xG^x{}_{x_4})+iG_{x_2x_4}(J_xG^x{}_{x_4})+iG_{x_3x_4}(J_xG^x{}_{x_2})-i(J_xG^x{}_{x_2})(J_xG^x{}_{x_3})(J_xG^x{}_{x_4})\big)Z[J]\Big|_{J=0}$$

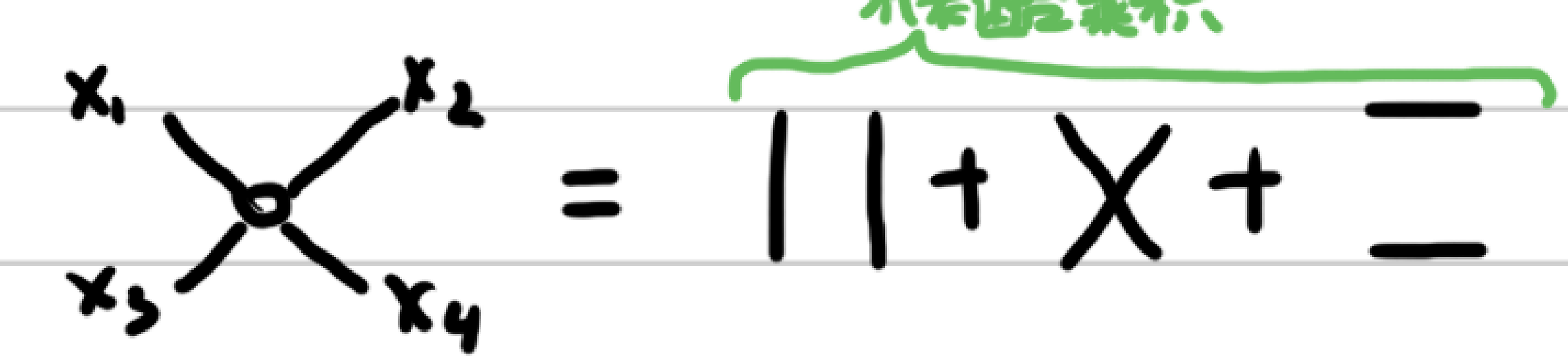
$$=G_{x_1x_2}G_{x_3x_4}+G_{x_1x_3}G_{x_2x_4}+G_{x_1x_4}G_{x_2x_3}$$

↓ 对点 G_2 组合乘积

Wick定理：n奇：$G_n=0$ n偶：$G_n=\langle G_{2k}\rangle$ → k对 G_2 组合乘积

Feynmann图

2点: 

4点: 

Feyman规则：图与公式的关系

不变函数

对 $\varphi^{(+)},\varphi^{(-)}$，有: $\langle 0|\varphi^{(+)}(x)\varphi^{(-)}(y)|0\rangle=\int\frac{d^3k\,d^3k'}{(2\pi)^3 2\sqrt{\omega\omega'}}\overbrace{\langle 0|a_k a^\dagger_{k'}|0\rangle}^{\delta_{kk'}(0+1)}e^{-ikx+ik'y}=\int\frac{d^3k}{(2\pi)^3 2\omega}e^{-ik(x-y)}=i\Delta_+(x-y)$

x,y换: $\langle 0|\varphi^{(+)}(y)\varphi^{(-)}(x)|0\rangle=\int\frac{d^3k}{(2\pi)^3 2\omega}e^{ik(x-y)}=i\Delta_-(x-y)=-i\Delta_+(y-x)$

↓

$$\Delta_F=\theta(x^0-y^0)\Delta_+(x-y)-\theta(y^0-x^0)\Delta_-(x-y)$$

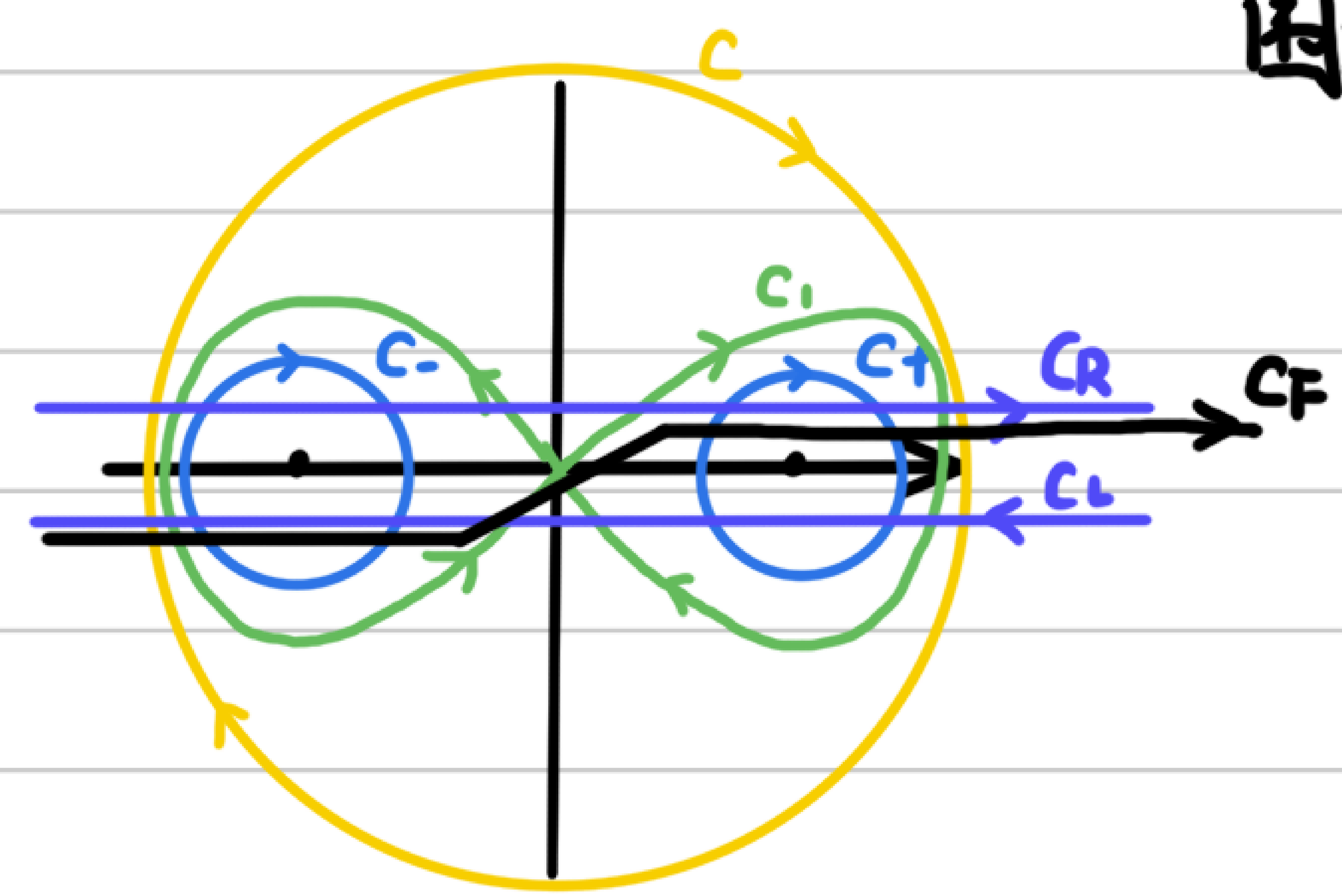
$$\Delta_\pm(x)=\int_{C_\pm}\frac{d^4k}{(2\pi)^4}\frac{1}{k^2-m^2+i\varepsilon}e^{-ikx}$$

对 $\int dk_0$ 每一项满足K-G方程 $(\partial_\mu\partial^\mu-m^2+i\varepsilon)\varphi=0$

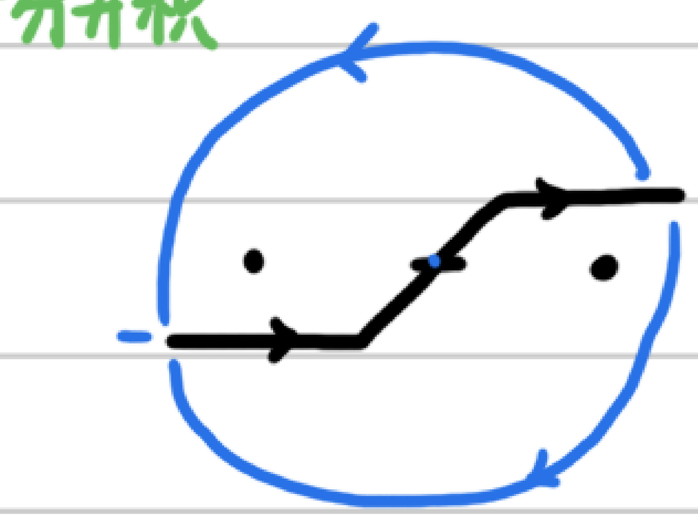
对 $k_0\in\mathbb{C}$，可取一围道 $C_\pm$

$$\Delta_F(x)=\int_{C_F}\frac{d^4k}{(2\pi)^4}\frac{1}{k^2-m^2+i\varepsilon}e^{-ikx}$$

围道图像



注意$\theta(x^0)$，上下两个分开积



$C_\pm$: $\Delta_\pm$的围道

C_F : Δ_F的围道，当$x^0>0$，$\overbrace{\theta(x^0)\Delta_+(x)}^{C_+}$为$C_F$+无穷下半圆

当$x^0<0$，$\underbrace{-\theta(-x^0)\Delta_-(x)}_{-C_-}$为$C_F$+无穷上半圆

Δ_F也叫因果Green函数Δ_c

C_R : Δ_R的围道，推迟Green函数，当$x^0>0$作无穷大下半圆，$\Delta_R=\theta(x^0)\Delta(x)$

C_A : Δ_A的围道，超前Green函数　$x^0<0$作无穷大上半圆，$\Delta_A=-\theta(-x^0)\Delta(x)$

这些Green函数的用途?

C : Δ的围道．$\Delta(x)=\Delta_+(x)+\Delta_-(x)\quad=\Delta_R-\Delta_A$

C_1 : Δ_1的围道．$\Delta_1(x)=i\int_{C_1}\frac{d^4k}{(2\pi)^4}\frac{e^{-ikx}}{k^2-m^2+i\varepsilon}=i[\Delta_+(x)-\Delta_-(x)]$

$\Delta,\ \Delta_\pm,\ \Delta_1$ 回路有限 → K-G方程 $(\partial_\mu\partial^\mu+m^2-i\varepsilon)\Delta=0$

$\Delta_F,\ \Delta_A,\ \Delta_R$ 回路无限 → $(\partial_\mu\partial^\mu+m^2-i\varepsilon)\Delta_F=-\delta(x-y)$

不同的回路代表不同边界条件 → 对应不同物理　　哪些不同物理

4. 旋量场路径积分

自由旋量场生成泛函　$\mathcal{L}_0=\bar\psi(i\gamma^\mu\partial_\mu-m)\psi$　ψ与$\bar\psi$为Grassmann变量

└ 外源与小虚数 : $\mathcal{L}=\mathcal{L}_0+\bar\eta\psi+\bar\psi\eta+i\varepsilon\bar\psi\psi=\bar\psi S_F^{-1}\psi+\bar\eta\psi+\bar\psi\eta$　(η与$\bar\eta$也为q数，才使得$\mathcal{L}$表现为c数)

跃迁振幅 : $K(q,t;q_0,t_0)=\int D\bar\psi D\psi\, e^{i\int dx\,\mathcal{L}(\psi,\bar\psi)}$　　$S_F^{-1}=i\gamma^\mu\partial_\mu-m+i\varepsilon=i\not\partial-m+i\varepsilon$

生成泛函 : $Z_0[\bar\eta,\eta]=\frac{1}{N}\int D\bar\psi D\psi\, e^{i\int dx\,\mathcal{L}(\psi,\bar\psi)}$

$=\frac{1}{N}\int D\bar\psi D\psi\, e^{i\int dx\,\bar\psi S_F^{-1}\psi+\bar\eta\psi+\bar\psi\eta}=e^{-i\bar\eta S_F\eta}$

$\left(N=\int D\bar\psi D\psi\, e^{i\int dx\,(\bar\psi+\bar\eta S_F)S_F^{-1}(\psi+S_F\eta)}\right)$

$$Z[\bar{\eta},\eta] = N\int D\bar{\psi}D\psi\, e^{i\int dx(\bar{\psi}S_F^{-1}\psi + \bar{\eta}\psi + \bar{\psi}\eta)}$$
$$= \langle 0|T e^{i\int dx(\bar{\eta}\psi + \bar{\psi}\eta)}|0\rangle$$

Dirac 传播子

$$\left.\begin{aligned}(i\partial\!\!\!/ - m + i\varepsilon)S_F &= \delta(x-y)\\ (i\partial\!\!\!/ - m + i\varepsilon)\psi &= -\eta\end{aligned}\right\} \longrightarrow \psi(x) = -\int dy\, S_F(x-y)\eta(y)$$

y:源 → x场

$$S_F(x) = \int \frac{d^4k}{(2\pi)^4} S_F(k) e^{-ikx} = \int \frac{d^4k}{(2\pi)^4}\frac{e^{-ikx}}{k\!\!\!/ - m + i\varepsilon}$$

ε作用为防0，故不留其它项

$$(k\!\!\!/ + m)(k\!\!\!/ - m + i\varepsilon) = \frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)k_\mu k_\nu - m^2 + i\varepsilon = k^2 - m^2 + i\varepsilon$$

$$S_F(x) = \int\frac{d^4k}{(2\pi)^4}\frac{k\!\!\!/ + m}{k^2 - m^2 + i\varepsilon}e^{-ikx} = (i\partial\!\!\!/ + m)\Delta_F$$

Green 函数：

Grassmann 交换反号

$$G(x,y) = \langle 0|T\psi(x)\bar{\psi}(y)|0\rangle, \quad T\psi(x)\bar{\psi}(y) = \theta(x^0-y^0)\psi(x)\bar{\psi}(y) - \theta(y^0-x^0)\bar{\psi}(y)\psi(x)$$

分离正负频：
$$\psi^{(+)}(x) = \int\frac{d^3k}{(2\pi)^3 2\omega}u(k,s)c_{ks}e^{-ikx} \qquad \bar{\psi}^{(-)}(x) = \int\frac{d^3k}{(2\pi)^3 2\omega}\bar{u}(k,s)c^\dagger_{ks}e^{ikx}$$
$$\psi^{(-)}(x) = \int\frac{d^3k}{(2\pi)^3 2\omega}v(k,s)d^\dagger_{ks}e^{ikx} \qquad \bar{\psi}^{(+)}(x) = \int\frac{d^3k}{(2\pi)^3 2\omega}\bar{v}(k,s)d_{ks}e^{-ikx}$$

$$G(x,y) = \theta(x^0-y^0)\langle 0|\psi^{(+)}(x)\bar{\psi}^{(-)}(y)|0\rangle - \theta(y^0-x^0)\langle 0|\bar{\psi}^{(+)}(y)\psi^{(-)}(x)|0\rangle$$

生成泛函导出：
$$G(x,y) = \left.\frac{-\delta^2 Z[\bar{\eta},\eta]}{i^2\delta\bar{\eta}(x)\delta\eta(y)}\right|_{\bar{\eta}=\eta=0} = iS_F(x-y)$$

为 $\frac{-\delta^2}{i^2\delta\bar{\eta}\delta\eta}$：$\frac{\delta}{i\delta\eta}$ 需与 $\bar{\psi}$ 对易才能作用于 η （或者说 $e^{\bar{\eta}S_F\eta}$ 中 $\frac{\delta}{\delta\bar{\eta}}$ 需与 η 对易）

有正反粒子之分（中间为负号）

以粒子传播方向为正　　$G(x,y) = y \longrightarrow x$